RACE
Regular Analysis Through Continuous Exercise

Liquid Solution : RACE-1

Introduction; Vapour Pressure; Henry's Law

1. Value of Henry's constant K_H _____.

(A) increases with increase in temperature.

(B) decreases with increase in temperature.

(C) remains constant.

(D) first increases then decreases.

CLQ235

2. Maximum amount of a solid solute that can be dissolved in a specified amount of a given liquid solvent does not depend upon _____.

(A) Temperature (B) Nature of solute

(C) Pressure (D) Nature of solvent

CLQ240

3. In the following questions a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

(A) Assertion and reason both are correct statements and reason is correct explanation for assertion.

(B) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

(C) Assertion is correct statement but reason is wrong statement.

(D) Assertion and reason both are incorrect statements.

(E) Assertion is wrong statement but reason is correct statement.

Assertion : When methyl alcohol is added to water, boiling point of water increases.

Reason : When a volatile solute is added to a volatile solvent elevation in boiling point is observed.

CLQ242

Raoult's Law

4. On the basis of information given below mark the correct option.

Information : On adding acetone to methanol some of the hydrogen bonds between methanol molecules break.

(A) At specific composition methanol-acetone mixture will form minimum boiling azeotrope and will show positive deviation from Raoult's law.

(B) At specific composition methanol-acetone mixture forms maximum boiling azeotrope and will show positive deviation from Raoult's law.

(C) At specific composition methanol-acetone mixture will form minimum boiling azeotrope and will show negative deviation from Raoult's law.

(D) At specific composition methanol-acetone mixture will form maximum boiling azeotrope and will show negative deviation from Raoult's law.

CLQ239

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5. Two completely miscible liquids A & B form an ideal solution. What is the vapour pressure (in torr) of solution containing 2 moles of A and 3 moles of B at 300K?

[Given: At 300 K, Vapour pr. of pure liquid A(P_A^o) = 100 torr

apour pr. of pure liquid B (P_B^o) = 300 torr]

CLQ243

Non-Ideal Solutions

6. On the basis of information given below mark the correct option. Information:

(i) In bromoethane and chloroethane mixture intermolecular interactions of A.A and B.B type are nearly same as A.B type interactions.

(ii) In ethanol and acetone mixture A.A or B.B type intermolecular interactions are stronger than A.B type interactions.

(iii) In chloroform and acetone mixture A.A or B.B type intermolecular interactions are weaker than A.B type interactions.

(A) Solution (ii) and (iii) will follow Raoult's law.

(B) Solution (i) will follow Raoult's law.

(C) Solution (ii) will show negative deviation from Raoult's law.

(D) Solution (iii) will show positive deviation from Raoult's law.

CLQ236

Distillation; Azeotrope

7. On dissolving sugar in water at room temperature solution feels cool to touch. Under which of the following cases dissolution of sugar will be most rapid?

(A) Sugar crystals in cold water.

(B) Sugar crystals in hot water.

(C) Powdered sugar in cold water.

(D) Powdered sugar in hot water.

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CLQ237

RLVP

8. Which of the following statements is false?

(A) Units of atmospheric pressure and osmotic pressure are the same.

(B) In reverse osmosis, solvent molecules move through a semipermeable membrane from a region of lower concentration of solute to a region of higher concentration.

(C) The value of molal depression constant depends on nature of solvent.

(D) Relative lowering of vapour pressure, is a dimensionless quantity.

CLQ241

Osmotic pressure

9. An unripe mango placed in a concentrated salt solution to prepare pickle, shrivels because _____.

(A) it gains water due to osmosis.

(B) it loses water due to reverse osmosis.

(C) it gains water due to reverse osmosis.

(D) it loses water due to osmosis.

CLQ234

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	
Answer	A	C	D	A	220	B	D	B	D	

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RACE
Regular Analysis Through Continuous Exercise

Liquid Solution : RACE-02

Raoult's Law

1. P_A = (235 y – 125 xy) mm of Hg. P_A is partial pressure of A, x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is the mole fraction of A in vapour phase, then $P°_B$ in mm of Hg is

(A) 235 (B) 0 (C) 110 (D) 125

CLQ214

RLVP

2. The vapour pressure of pure benzene at 50°C is 268 torr. How many mol of non-volatile solute per mol of benzene is required to prepare a solution of benzene having a vapour pressure of 167 torr at 50°C:

(A) 0.377 (B) 0.605 (C) 0.623 (D) 0.395

CLQ215

3. Vapour pressure of an aqueous solution of a non-volatile and non electrolyte solute at certain temperature T was found to be 39.964 torr. If the vapour pressure of pure solvent at same temperature is 40 torr, find the molality of aqueous solution:

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(A) 0.1 m (B) 0.005 m (C) 0.05 m (D) 0.04 m

CLQ217

Depression in freezing point

4. **Statement-1** : The freezing of water is an endothermic process.

Statement-2 : Heat must be removed from the water to make it freeze.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CLQ222

Osmotic pressure

5. Solutions having the same osmotic pressure are called :

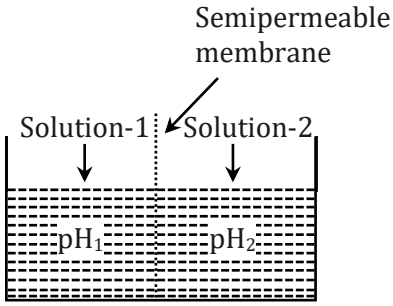
(A) isotonic solution (B) molar solutions

(C) hypotonic solutions (D) ideal solutions

CLQ218

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6.



If water particles move from solution-1 to solution-2 when both the aqueous solution contain same base at same temperature then

(A) $pH_1 > pH_2$ (B) $pH_1 < pH_2$ (C) $pH_1 = pH_2$ (D) $pH_1 \geq pH_2$

CLQ219

Abnormal Colligative Properties, Van't Hoff Factor

7. 1.0 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is ($K_{b(H_2O)}$ = 0.52 K kg mol^{-1})

(A) 274.76 K (B) 377 K (C) 376.4 K (D) 374.76 K

CLQ220

8. What will be the molecular weight of $CaCl_2$ determined in its aq. solution experimentally from depression of freezing point ?

(A) 111 (B) <111 (C) > 111 (D) data insufficient

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CLQ221

9. The vapour pressure of a dilute solution of a solute is not influenced by :

(A) temperature of solution (B) melting point of solute

(C) mole fraction of solute (D) degree of dissociation of solute

CLQ216

10. Calculate the molality of aq. NaCl solution if boiling point is 100.10°C. [$K_b(H_2O)$ = 0.5 K kg mol^{-1}].

Fill your answer by multiplying it with 10

CLQ223

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	C	D	A	B	D	B	B	1

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Regular Analysis Through Continuous Exercise

Liquid Solution : RACE-03

Increase in boiling point

1. Elevation of boiling point of 1 molar aqueous glucose solution (density = 1.2 g/ml) is

(A) K_b (B) 1.20 K_b (C) 1.02 K_b (D) 0.98 K_b

CLQ229

2. A 2% (w/v) solution of glucose has the same elevation in B.P. as that of 5% (w/v) solution of a non-volatile solute the molar mass of the solute is -

(A) 180 (B) 450 (C) 72 (D) 18

CLQ232

Depression in freezing point

3. How many grams of sucrose (mol. wt. = 342) should be dissolved in 100 gm water in order to produce a solution with 105° C difference between the freezing point & boiling point temperatures?

(Unit : k_f = 2 K.kg mol^{-1};k_b = 0.56 K.kg mol^{-1})

(A) 34 gm (B) 46 gm (C) 67 gm (D) 342 gm

CLQ230

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4. For a liquid normal boiling point is -173°C then at 2 atm pressure it's boiling point should be nearly

($\Delta H_{vap.}$ = 200 cal/mole, R=2 cal/mol-KI, ln 2 = 0.7)

(A) –73°C (B) 333°C (C) 60°C (D) 103°C

CLQ231

Abnormal Colligative Properties, Van't Hoff Factor

5. A 2% (w/v) solution of glucose has the same elevation in B.P. as that of 5% (w/v) solution of a non-volatile solute the molar mass of the solute is -

(A) 180 (B) 450 (C) 72 (D) 18

CLQ224

6. A solution prepared by dissolving a 2.50 g sample of an unknown compound dissolved in 34.0 g of benzene, C_6H_6 boils at 1.38°C higher than pure benzene. Which expression gives the molar mass of the unknown compound ?

Compound	**K_b**
C_6H_6	2.53°C.m^{-1}

(A) $2.53\times\frac{2.50}{1.38}$ (B) $1.38\times\frac{34.0}{2.53}\times 2.50$

(C) $2.50\times10^3\times\frac{2.53}{34.0}\times\frac{1}{1.38}$ (D) $2.50\times10^3\times\frac{1.38}{34.0}\times 2.53$

CLQ225

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7. When 1.20 g of sulphur is melted with 15.00 g of naphthalene, the solution freezes at 77.2° C. What is the molar mass of this from of sulphur. Data for Napthalene

Melting point, m.p 80°C

Freezing point depression constant, $k_f = 6.80°C\ m^{-1}$

(A) 180 g mol^{-1} (B) 194 g mol^{-1} (C) 260 g mol^{-1} (D) 450 g mol^{-1}

CLQ226

8. The Van't Hoff factor 0.1 M La $(NO_3)_3$ solution is found to be 2.74 the percentage dissociation of the salt is :

(A) 85 % (B) 58 % (C) 65.8% (D) 56.8%

CLQ227

9. Maximum freezing point will be for 1 molal solution of (assuming equal ionisation in each case):

(A) $[Fe(H_2O)_3\ Cl_3]$ (B) $[Fe(H_2O)_5Cl]\ Cl_2.H_2O$

(C) $[Fe(H_2O)_4Cl_2]Cl.2H_2O$ (D) $[Fe(H_2O)_3Cl_3].3H_2O$

CLQ228

10. Equal volumes of 1.0 M KCl and 1.0 M $AgNO_3$ are mixed. The depression of freezing point of the resulting solution will be ($K_f(H_2O)$ = 1.86 K kg mol^{-1}, 1 M = 1m)

CLQ233

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	C	B	C	B	B	D	1.86

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